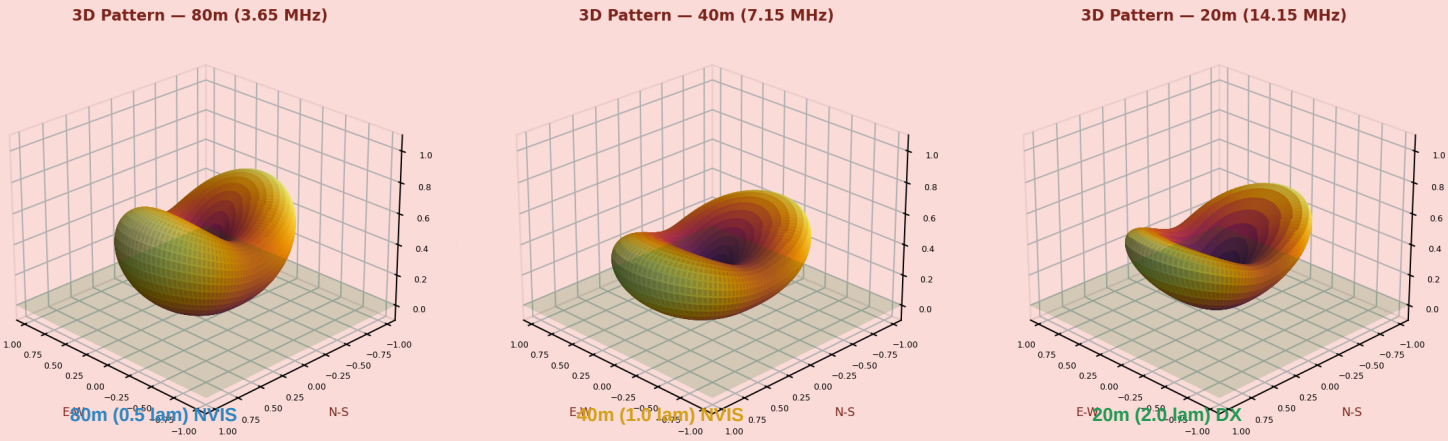
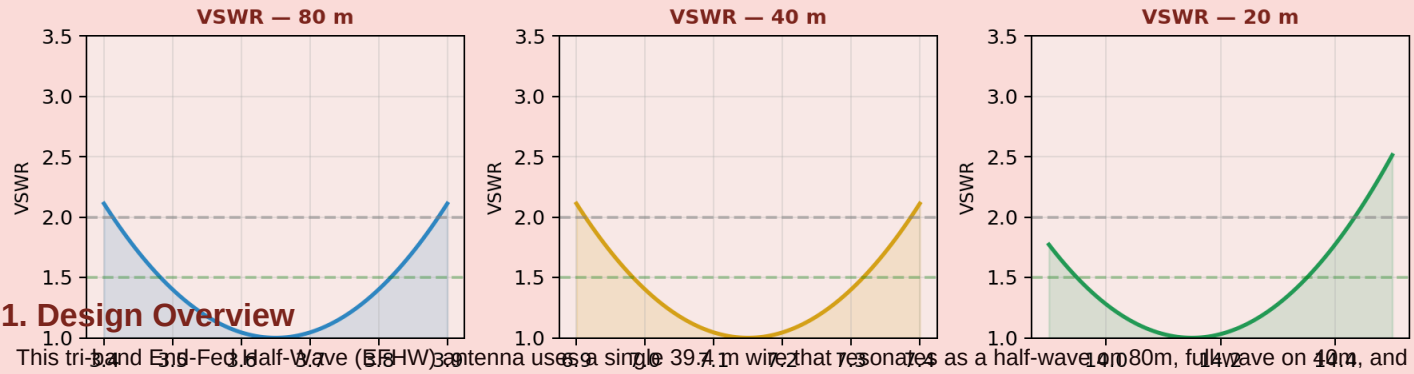
# Tri-Band End-Fed Half-Wave (EFHW)

80 m + 40 m + 20 m | Inverted-L | 49:1 Transformer

*General HF: NVIS on 80m/40m + low-angle DX on 20m from a single wire*




## 1. Design Overview

This tri-band End-Fed Half-Wave (EFHW) antenna uses a single 39.4 m wire that resonates as a half-wave on 80m, full-wave on 40m, and double-wave on 20m through natural harmonic relationship. A 49:1 impedance transformer at the feed end converts the ~2500 ohm end-impedance to 50 ohm for direct coax feed.

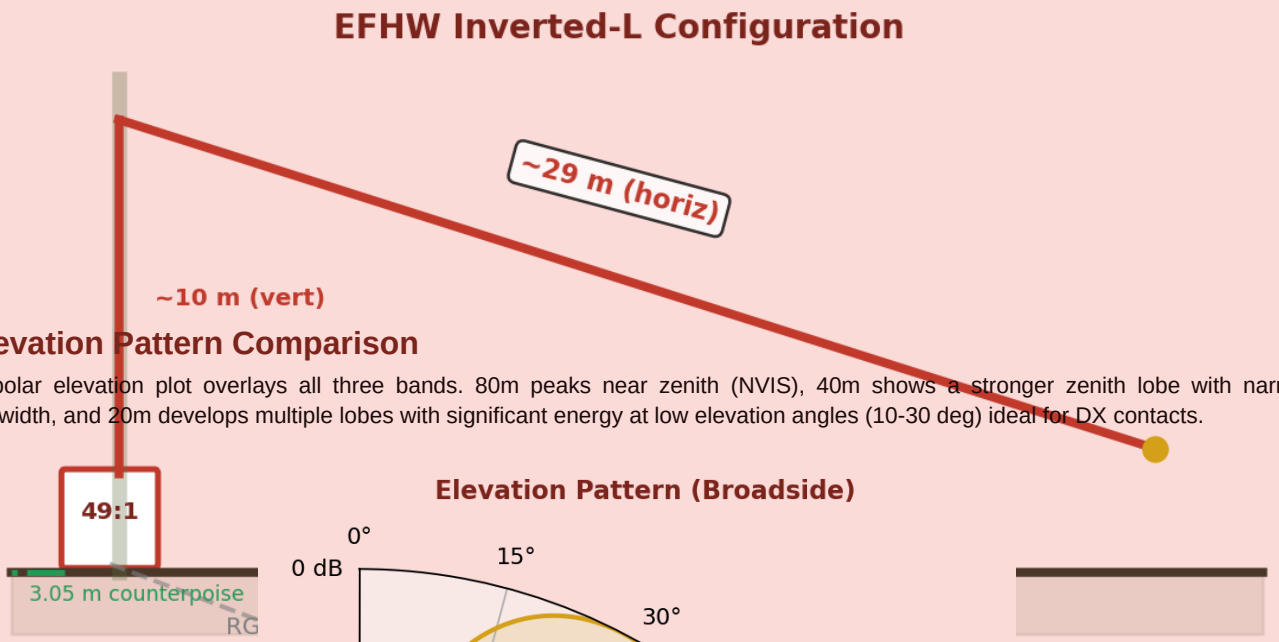
The inverted-L configuration places the transformer at ground level with the wire running vertically up a 10-12 m support, then sloping down to a far-end anchor at 2-5 m. This provides both NVIS coverage on 80m/40m (high-angle radiation) and DX capability on 20m (low-angle radiation from the 2-lambda pattern).

Unlike trap antennas, the EFHW achieves multi-band operation with no traps, loading coils, or antenna tuner, keeping the system simple, efficient, and inexpensive.

### Key Parameters

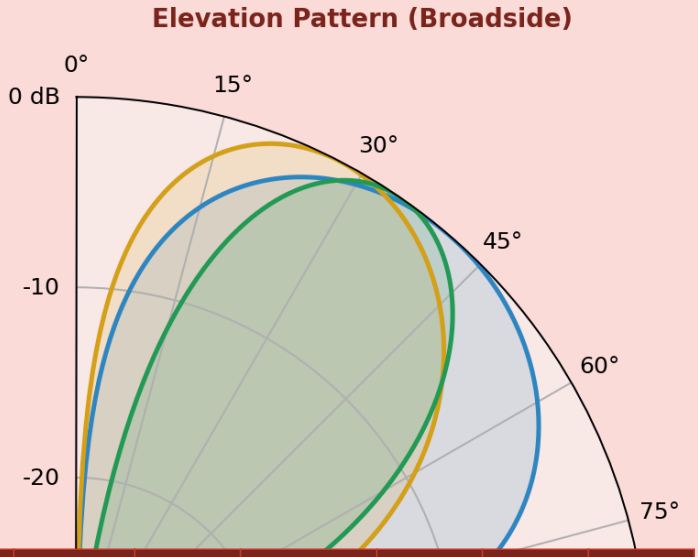
| | |
|---|---|
| **Type:** | End-Fed Half-Wave (EFHW), inverted-L |
| **Bands:** | 80m (3.5-3.8) + 40m (7.0-7.3) + 20m (14.0-14.35 MHz) |
| **Wire Length:** | 39.4 m, #14 AWG stranded copper |
| **Vertical Section:** | ~10 m (up the support) |
| **Horizontal Section:** | ~29 m (sloping to far end at 2-5 m) |
| **Transformer:** | 49:1 on 2x FT-240-43 (2:14 turns) |
| **Counterpoise:** | 3.05 m wire on ground |
| **Feed Cable:** | RG-213 (50 ohm) with CM choke |
| **Power:** | 200 W PEP (80m), 300 W (40m/20m) |
| **Cost:** | $95-$185 |

## 2. Antenna Configuration




## 3. Elevation Pattern Comparison

The polar elevation plot overlays all three bands. 80m peaks near zenith (NVIS), 40m shows a stronger zenith lobe with narrower beamwidth, and 20m develops multiple lobes with significant energy at low elevation angles (10-30 deg) ideal for DX contacts.




## 4. Performance Data

| Freq | Band | Mult | Gain | Eff% | VSWR | BW | h/lam | Use |
|---|---|---|---|---|---|---|---|---|
| 3.500 | 80m | 0.5 lam | +2.0 | 85 | 2.0:1 | 100 | 0.12 | NVIS |
| 3.650 | 80m | 0.5 lam | +2.8 | 88 | 1.5:1 | 120 | 0.12 | NVIS |
| 3.800 | 80m | 0.5 lam | +3.0 | 88 | 2.2:1 | 120 | 0.13 | NVIS |
| 7.000 | 40m | 1.0 lam | +4.5 | 92 | 1.5:1 | 200 | 0.23 | NVIS |
| 7.150 | 40m | 1.0 lam | +5.0 | 93 | 1.2:1 | 220 | 0.24 | NVIS |
| 7.300 | 40m | 1.0 lam | +5.2 | 93 | 1.6:1 | 220 | 0.24 | NVIS |
| 14.000 | 20m | 2.0 lam | +6.5 | 95 | 1.8:1 | 250 | 0.47 | DX |
| 14.150 | 20m | 2.0 lam | +7.0 | 96 | 1.5:1 | 280 | 0.47 | DX |
| 14.350 | 20m | 2.0 lam | +7.2 | 96 | 1.8:1 | 280 | 0.48 | DX |

## 5. EFHW Operating Principle

The EFHW exploits the harmonic relationship between amateur bands: 3.5 MHz x 2 = 7 MHz x 2 = 14 MHz. A wire that is lambda/2 at 3.5 MHz is automatically 1 lambda at 7 MHz and 2 lambda at 14 MHz. At each harmonic, the feed-point impedance remains high (~2500-5000 ohm), so the same 49:1 transformer provides a reasonable match on all three bands.

On 80m (0.5 lambda), the pattern is a classic half-wave broadside lobe peaking at zenith - ideal for NVIS. On 40m (1.0 lambda), the pattern narrows with higher gain still peaking upward. On 20m (2.0 lambda), multiple lobes form with significant energy at lower elevation angles (20-40 deg), providing DX capability that the lower bands cannot achieve.

## 6. 49:1 Transformer Construction

Stack two FT-240-43 toroids and tape together. Wind 14 turns of #14 AWG PTFE-insulated wire as the secondary (antenna side), then wind 2 turns as the primary (coax side). The impedance ratio is (14/2)^2 = 49:1, transforming ~2500 ohm to ~51 ohm. House in an IP65 ABS enclosure with an SO-239 connector for the coax and a wing-nut terminal for the antenna wire. Add a second terminal for the counterpoise.

| Parameter | Value | Notes |
|---|---|---|
| **Core** | 2x FT-240-43 stacked | Fair-Rite 43 NiZn ferrite |
| **Primary** | 2 turns #14 AWG PTFE | Coax side (50 ohm) |
| **Secondary** | 14 turns #14 AWG PTFE | Antenna side (~2500 ohm) |
| **Ratio** | (14/2)^2 = 49:1 | 2500 -> 51 ohm |
| **Loss (80m)** | 0.3-0.5 dB | Lower ferrite mu at 3.5 MHz |
| **Loss (40/20m)** | 0.1-0.3 dB | Better efficiency at higher f |
| **Power** | 200W (80m) / 300W (40/20m) | Core saturation limit on 80m |
| **Housing** | IP65 ABS enclosure | SO-239 + wing-nut terminals |

## 7. Bill of Materials

| # | Item | Qty | Cost (USD) | Notes |
|---|---|---|---|---|
| 1 | #14 AWG stranded Cu wire, 42 m | 1 | $10-20 | THHN or bare; +trim allowance |
| 2 | FT-240-43 ferrite toroid | 2 | $20-30 | For 49:1 transformer |
| 3 | #14 AWG PTFE wire (transformer) | 2 m | $3-5 | For primary + secondary |
| 4 | RG-213 coaxial cable, 20 m | 1 | $25-40 | 50 ohm, UV resistant |
| 5 | FT-240-43 (CM choke) | 1 | $10-15 | 8-10 turns RG-213 |
| 6 | IP65 ABS enclosure | 1 | $5-10 | For transformer housing |
| 7 | SO-239 chassis connector | 2 | $3-5 | Transformer + choke |
| 8 | Mast/support hardware | 1 | $15-50 | Fibreglass push-up or tree |
| 9 | Dacron rope, insulators, misc | 1 | $5-10 | UV-resistant support line |
| | **TOTAL** | | **$95-$185** | |

## 8. Comparison with Other Designs

| Parameter | EFHW Tri-Band (this design) | Efficient Fan Dipole | Small-Size Fan Dipole | Balanced Mag Loop |
|---|---|---|---|---|
| **Bands** | **80+40+20m** | 80+40m | 80+40m | 80+40m |
| **80m Gain** | **+2.8 dBi** | +7.0 dBi | +2.5 dBi | -7 dBi |
| **40m Gain** | **+5.0 dBi** | +7.8 dBi | +5.5 dBi | +1 dBi |
| **20m Gain** | **+7.0 dBi** | N/A | N/A | N/A |
| **Efficiency** | **85-96%** | 97-99% | 72-86% | 8-55% |
| **Footprint** | **29x10 m** | 39x12 m | 19x8 m | 2 m dia |
| **Supports** | **1 mast** | 1 mast | 1 mast | Tripod |
| **Cost** | **$95-185** | $280-450 | $80-150 | $313-712 |

## 9. Tuning Procedure

1. Cut wire to 40.5 m (includes 1.1 m trim allowance).
2. Connect to 49:1 transformer. Attach 3.05 m counterpoise.
3. Erect in inverted-L configuration (~10 m vertical + ~30.5 m sloping).
4. Sweep 3.4-3.9 MHz with analyser: find VSWR minimum. Target: 3.65 MHz.
5. Sweep 7.0-7.3 MHz: should show VSWR < 1.6:1 near 7.15 MHz.
6. Sweep 14.0-14.35 MHz: should show VSWR < 2.0:1 near 14.15 MHz.
7. If all bands are high: trim 20 cm at a time from the far end.
8. If all bands are low: wire is too short; splice and extend.
9. Fine-tune: adjust counterpoise length +/- 10 cm for best 80m match.
10. Install CM choke (8-10 turns RG-213 on FT-240-43) near transformer.

**QUICK REFERENCE CARD**

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TYPE: EFHW Inverted-L  |  BANDS: 80m+40m+20m  |  WIRE: 39.4m #14 AWG  |  XFMR: 49:1 (2x FT-240-43)
80m: +2.8 dBi, 88% eff, BW 120 kHz, NVIS  |  40m: +5.0 dBi, 93% eff, BW 220 kHz, NVIS
20m: +7.0 dBi, 96% eff, BW 280 kHz, DX    |  Feed: RG-213 50 ohm  |  Cost: $95-$185
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